FEASIBILITY STUDY

FloodSense: An Augmented Reality Mobile Application with Machine

Learning-Based Flood Susceptibility Mapping for Bacoor City

Presented to

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OVERVIEW

Flooding remains one of the most persistent and consequential natural hazards affecting Bacoor City, Cavite. Located in a low-lying coastal area within Cavite province, Bacoor City is regularly exposed to flooding driven by monsoon rainfall, storm surges, and inadequate drainage infrastructure. As supported by the findings of the primary data gathering phase of this study, the impact of flooding on residents is not only physical but also informational — residents consistently lack timely, accurate, and location-specific flood susceptibility information that would allow them to make informed decisions before and during flood events.

A structured survey administered to 50 residents across 12 flood-prone barangays in Bacoor City revealed significant gaps in the current state of flood information dissemination. Of all respondents, 92% reported experiencing flooding in their area, with 54% experiencing it almost every rainy season and 58% reporting severe flooding. Despite the frequency and severity of flooding, only 20% of respondents considered themselves well-informed about flood-prone areas in Bacoor City — with the remaining 80% falling under "Somewhat Aware," "Not Sure," or "Not Aware at All." The three Likert-scale items on flood information quality (Q9–Q11) yielded mean scores of 2.62, 2.70, and 2.72 respectively, indicating near-dissatisfied perceptions of current information quality, timeliness, and clarity. Most critically, 64% of respondents — 32 out of 50 — reported having made a wrong decision due to insufficient flood information, including taking flooded roads, failing to evacuate in time, and misjudging flood conditions.

Thematic analysis of open-ended survey responses further confirmed the nature of this gap. The two most prevalent themes from Q12 — "Biggest Problem in Getting Flood Information" — were Late or Delayed Information (26%) and Lack of Information or No Warnings at All (20%), together accounting for 38% of all responses. Respondents also cited unclear or inaccurate information (12%) and lack of location-specific flood susceptibility information (6%) as recurring barriers. The survey findings were subsequently validated through a Key Informant Interview with a staff member of the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), who confirmed that the office's current flood information dissemination relies primarily on social media, email, and barangay coordination — and acknowledged that resident engagement with official warning channels remains low due to poverty, work constraints, and low attendance at flood orientations.

The BDRRMO interview further revealed that while the office maintains historical flood data and GIS-based exposure maps per barangay, these datasets are not accessible to the general public in a usable or interactive format. The interviewee confirmed that the office's data holdings include barangay boundary maps, historical flood records (affected areas and flood events), and field-assessed exposure data — and expressed openness to data sharing for research and academic purposes. Critically, the BDRRMO officer affirmed that a mobile-based tool providing location-specific flood susceptibility information would serve a genuine community need — particularly for residents in **highly susceptible barangays** who require actionable flood awareness support for area-avoidance and evacuation decisions.

A supplementary problem-identification interview conducted on May 18, 2026 with the Officer-in-Charge of the Bacoor City Planning and Development Office (CPDCO) confirmed that the office holds land-use maps, elevation maps, flood-hazard maps, and barangay boundary maps, and that these can be released to the research team as hard-copy prints using the existing endorsement letter. The CPDCO officer further confirmed that Bacoor functions as a regional catch basin/low-lying area and that drainage degradation is a flood driver, validating the environmental features selected for the susceptibility model.

FloodSense is a proposed mobile application that addresses these identified gaps through the integration of machine learning-based flood susceptibility classification and augmented reality visualization. Using a supervised ML classification model trained on environmental features — including historical flood records, elevation data, and rainfall parameters — the system classifies **flood susceptibility** across Bacoor City's barangays into four levels: Low, Moderate, High, and Very High. These classifications are rendered through an augmented reality ground-plane overlay, allowing users to visualize susceptibility levels by pointing their phone camera at any flat surface. Rainfall input is simplified through a PAGASA-aligned dropdown menu using plain-language intensity presets (Light, Moderate, Heavy, Intense and Torrential), ensuring usability for non-technical community users. The system targets all residents Bacoor City — providing a decision support tool that bridges the gap between institutional flood data and actionable community-level flood awareness.

KEY FEATURES

Core Features

1. Augmented Reality Flood Susceptibility Map

Renders a color-coded flood susceptibility map (Low/Moderate/High/Very High) per barangay through the phone camera using Vuforia ground-plane markerless tracking. The AR overlay is **anchored to a detected flat surface** — not GPS-linked to the user's outdoor position. A surface detection indicator (green = valid, red = invalid) ensures accurate rendering before the map is displayed. The map shows a Legend overlay only; title, date, and data source are accessible through an in-app info panel to avoid AR clutter.

- Color-coded susceptibility levels: Low (green), Moderate (yellow), High (orange), Very High (red)
- Markerless ground-plane tracking via Vuforia — works on any flat horizontal surface (table, floor)
- Surface detection indicator — green/red; map inputs locked until surface confirmed valid
- Legend visible in AR view; full metadata (STANDL elements) in info panel
- Disclaimer accessible from AR view: susceptibility levels indicate physical flood likelihood, not a real-time guarantee

2. PAGASA-Aligned Rainfall Input

Users select rainfall scenario and duration from dropdown menus using plain-language labels tied to PAGASA's public rainfall advisory classification. The app internally maps these selections to millimeter-per-hour values used by the ML model removing the need for technical numeric input from non-expert users. This design was directly validated by the BDRRMO key informant.

- Rainfall intensity presets: Light (<2.5 mm/hr), Moderate (2.5– 7.5 mm/hr), Heavy (7.5 - 15 mm/hr), Intense(15 - 30mm/hr), Torrential (<30 mm/hr)
- Duration dropdown: 1 hr, 3 hrs, 6 hrs, 12 hrs, 24 hrs
- Rain intensity icons alongside text labels (optional)
- Reference: PAGASA Rainfall Advisory System

3. Machine Learning Flood Susceptibility Classification

A supervised classification model trained on environmental datasets classifies flood susceptibility per barangay based on user-selected rainfall parameters combined with pre-loaded spatial features. The model outputs one of four susceptibility levels per barangay zone, which directly drives the AR map color coding. The key spatial feature, elevation, is pre-identified by the research team from institutional maps — not collected from community surveys.

- Classification output: Low / Moderate / High / Very High
- Features: elevation, historical flood records, rainfall intensity and duration
- Data sources: BDRRMO historical flood records, CPDCO hard-copy maps (elevation, land use, hazard) supplemented by open digital elevation models, and PAGASA rainfall classification tables
- Model: Random Forest or XGBoost (scikit-learn, Python)

4. User Account and Preference System

Users create an account to access personalized features including saved rainfall preferences. Firebase Authentication handles user management with secure credential storage.

- Email/password account creation and login
- Saved rainfall preference parameters (intensity and duration)
- User profile and settings management

5. Onboarding and Disclaimer Screen

A first-launch onboarding screen informs users of the app's purpose, limitations, and proper usage — ensuring informed use and reducing the risk of misinterpreting susceptibility outputs as real-time flood guarantees.

- App purpose and scope explanation
- Disclaimer: 'This app models rainfall-induced flood susceptibility. It is not a real-time flood sensor. Always follow official warnings from BDRRMO and PAGASA.'
- Flat surface instruction for AR usage

I. TECHNICAL FEASIBILITY

This section evaluates whether FloodSense can be built using currently available technologies, tools, and frameworks within the capability of an undergraduate Computer Science team. Each component is mapped to a specific, free or low-cost technology with a documented implementation path.

1.1 Proposed Technology Stack

Layer	Technology	Cost	Justification
Mobile App (all screens)	Unity + uGUI (Canvas, Input Fields, Dropdowns)	Free (Personal License)	Unified development environment for both UI screens and AR; eliminates Flutter/React Native handoff
AR Engine	Vuforia SDK (Ground Plane)	Free tier	Markerless ground-plane tracking on flat surfaces; no GPS anchoring required
Backend / Database	Firebase (Firestore, Auth)	Free tier	Real-time sync, built-in authentication, cloud storage
ML Model	Python (pandas, NumPy, scikit-learn, XGBoost)	Free	Flood susceptibility classification; trained on open and BDRRMO datasets
ML Output Storage	Firebase Firestore (pre-computed JSON lookups)	Free	The trained model generates susceptibility lookup tables (JSON) for all supported rainfall intensity/duration combinations. These static files are stored in Firestore. The Unity app fetches the appropriate document directly
Elevation / Spatial Data	CPDCO (via data-sharing agreement, collaboration)	Free (institutional	Topographic and land use features for ML model training
Historical Flood Data	BDRRMO (via data-sharing agreement)	Free (institutional)	Ground-truth flood occurrence data per barangay
Rainfall Reference	PAGASA Advisory Classification	Free (public)	Basis for user-facing rainfall intensity presets
UI/UX Prototyping	Figma	Free tier	Mobile interface design and prototyping prior to Unity development

1.2 Feature-Level Technical Assessment

Feature	Implementation Approach	Complexity	Est. Time
AR Flood Susceptibility Map (Vuforia Ground Plane)	Unity + Vuforia ground-plane detection; flat-surface anchored barangay zone overlay; color-coded susceptibility rendering	High	4 weeks
PAGASA Rainfall Dropdown + Susceptibility Lookup	Unity uGUI Dropdown mapped to mm/hr values; app reads the matching pre-computed susceptibility JSON document from Firestore (keyed by intensity + duration	Medium	2 weeks
ML Classification Model (Python)	scikit-learn/XGBoost trained on elevation, historical flood records, and rainfall features. The model is used offline to generate pre-computed susceptibility lookup tables (JSON) for all supported rainfall intensity/duration combinations. These tables are uploaded to Firestore. No live model inference is required during app runtime.	High	4 weeks
Results Summary Screen	Unity uGUI ScrollView / List of barangay names + color badge; fed directly from ML response stored in Firestore	Low	1 week
User Account + Auth (Firebase)	Firebase Authentication for email/password login; Firestore for user preferences	Low	1 week
Surface Detection Indicator	Vuforia ground plane status callback; tint material green/red based on surface validity; lock input until green	Low	1 week
Onboarding + Disclaimer Screen	Unity first-launch scene with instructions and disclaimer; dismissable with user acknowledgement	Low	0.5 week
Flood Information Hub (stretch)	Static Firestore-backed content screen; barangay flood history, evacuation info, BDRRMO contacts	Low	1 week
Offline Mode (stretch)	Locally cached ML model and barangay data; Firebase offline persistence for user data	Medium	2 weeks

1.3 Technical Challenges and Mitigations

AR Rendering Performance on Mid-Range Devices

Vuforia ground-plane AR and barangay zone overlay rendering may cause performance issues on entry-level Android devices.

Mitigation: Optimize barangay zone geometry by simplifying polygon vertices. Implement level-of-detail rendering. Set minimum device specification in app description.

User Input Accuracy for Rainfall Selection

Users may not correctly identify rainfall intensity without reference, leading to inaccurate model queries.

Mitigation: Use PAGASA-aligned plain-language labels (Light, Moderate, Heavy, Intense, and Torrential) with optional rain intensity icons. Add brief instruction: 'Select the rainfall level that best matches current or expected conditions.' Directly validated by the BDRRMO key informant.

Flat Surface Detection Edge Cases

Users may point the camera at non-flat surfaces (grass, fabric, uneven floors), preventing ground-plane detection.

Mitigation: Vuforia's Ground Plane automatically detects surface validity. Implement green/red surface indicator and a persistent instruction message: 'Point camera at a flat table or floor.' Block input form from appearing until surface is confirmed green.

II. OPERATIONAL FEASIBILITY

This section evaluates whether FloodSense can be realistically developed, deployed, and used within the constraints of the thesis timeline, team size, target users, and available institutional resources.

2.1 Team Composition and Role Distribution

Member	Primary Role	Responsibilities
Member 1	Mobile / AR Developer	Unity uGUI screen development (Login, Preferences, Results Summary, AR View), Unity scene navigation, user account UI, onboarding flow, Vuforia AR integration, surface detection UI
Member 2	ML Engineer / Backend	Python ML model development (feature engineering, training, evaluation), Firebase setup, ML model, Firestore integration
Member 3	Data / Research Lead	Spatial data collection and preprocessing (CPDCO, BDRRMO), barangay zone preparation, usability testing coordination, thesis documentation

2.2 Recommended Development Timeline (7-10 Months)

Phase	Duration	Deliverables
Phase 1 — Foundation	Months 1–2	Project setup, Firebase configuration, user authentication (email/password), basic Unity UI screens (Login, Preferences placeholder, Results Summary placeholder), formal data request to BDRRMO/CPDCO, initial data preprocessing (cleaning, feature engineering).
Phase 2 — ML Model Development	Months 2–4	Data cleaning and feature engineering using pandas and NumPy (elevation, historical flood records, rainfall parameters). Training and evaluation of Random Forest / XGBoost classifier (scikit-learn). Generation of pre-computed susceptibility lookup tables (JSON) for all supported rainfall intensity/duration combinations. Upload of JSON files to Firestore
Phase 3 — AR Integration	Months 4–6	Vuforia Unity ground-plane AR setup; flat-surface barangay zone overlay; color-coded susceptibility rendering; surface detection indicator; legend overlay

Post-Development Phase (Months 8–10)

Phase	Activities
Months 8–9	Usability testing with at least 20 residents from survey pool, BDRRMO review of output, bug fixes, ML model evaluation metrics write-up, final refinements.

Phase	Activities
Month 10	Final defense preparation, documentation completion, thesis manuscript finalization.

2.3 Target Users and Adoption Feasibility

User Tier	Description	Adoption Strategy
Tier 1 — Bacoor City Residents (flood-prone barangays)	Residents of historically susceptible barangays (Zapote, Mambog, Talaba, Niog) who experience flooding regularly and need location-specific susceptibility information for evacuation and safety decisions	Pilot distribution through BDRRMO community outreach and barangay-level orientation; direct recruitment from survey respondents
Tier 2 — General Bacoor Residents	Residents outside the highest-susceptibility barangay zones who use the app for general flood awareness and community preparedness planning	App store distribution; awareness through city-level flood information campaigns

2.4 Financial Feasibility

FloodSense is designed to operate at zero licensing cost during the thesis development and pilot testing phases. All tools listed operate within free tiers sufficient for a thesis-scale pilot (20–50 participants).

Tool / Service	Purpose	Cost
Unity + Vuforia SDK	AR development, ground-plane tracking, and all UI screens (uGUI)	Free (Personal License)
Firebase Spark Plan	Auth, Firestore database, Cloud Functions	Free tier
Python / scikit-learn / XGBoost	ML model training and evaluation	Free
CPDCO Topographic Data	Elevation features for ML model	Free (data-sharing agreement)
BDRRMO	Flood hazard maps and rainfall data	Free (open access)
PAGASA Rainfall Advisory Data	Rainfall intensity classification reference	Free (public)
BDRRMO Historical Flood Data	Ground-truth barangay flood records	Free (data-sharing agreement)
Figma	UI/UX prototyping	Free tier

2.5 Operational Risks and Mitigations

BDRRMO Data Access Delay

BDRRMO may require additional time to process a formal data-sharing request, potentially delaying ML model development.

Mitigation: Initiate formal data request immediately after title defense approval. Begin ML model development in parallel using PAGASA open datasets as primary sources. Treat BDRRMO data as supplementary validation input to avoid single-source dependency.

CPDCO Spatial Data Inaccessibility

CPDCO may not have digital GIS files and shapefiles readily available, or may store data in formats incompatible with the research pipeline.

Mitigation: Hard-copy maps can be georeferenced and digitised using QGIS, converting them into usable vector or raster layers. If digitisation time exceeds the project schedule, open digital elevation models (SRTM, ALOS PALSAR) and NAMRIA land-cover data will be used directly. All data sources and processing steps will be documented in the methodology.

Low Pilot User Adoption

Community residents, particularly elderly or low-literacy users, may find the AR interface unfamiliar or difficult to use.

Mitigation: Conduct usability testing with at least 20 residents from the original survey pool. Simplify onboarding with visual instructions. Leverage BDRRMO community outreach as a distribution channel for pilot recruitment.

Scope Creep — AR + ML Dual Complexity

Developing both an AR system (Unity/Vuforia) and an ML pipeline (Python) in parallel is technically demanding for a thesis team.

Mitigation: Assign dedicated roles — AR/UI development and ML development handled by separate team members. Phase timeline separates ML completion (Phase 2) from AR integration (Phase 3) to prevent bottlenecks. Offline mode and Flood Information Hub are stretch goals deferred to Phase 4 only if earlier phases are completed ahead of schedule.

Data Privacy Compliance

User account data is subject to Philippine Republic Act 10173 (Data Privacy Act).

Mitigation: Firebase Security Rules enforce per-user data isolation. Informed consent obtained from all pilot participants. No personal location data is collected, stored, or transmitted. The AR overlay is anchored to a detected flat surface — the user's GPS coordinates are not used or stored by the application.

Sustainability Post-Thesis

Without a formal data-sharing agreement, the app's ML model may become outdated as new flood events occur.

Mitigation: Frame sustainability as future work in the thesis defense. Proposed sustainability pathway: establish a formal MOU with BDRRMO for periodic model retraining using updated historical flood records, and explore integration with BDRRMO's existing community information dissemination channels.

Interview-Identified Gap: User Input Accuracy

As raised by the BDRRMO key informant, users cannot accurately estimate rainfall in millimeters, making free-text or numeric input unreliable.

Mitigation: Fully addressed through the PAGASA-aligned dropdown design — users select plain-language intensity levels (Light, Moderate, Heavy, Intense, and Torrential) tied to standardized mm/hr ranges. The app handles all technical conversion internally. This design decision was directly validated by the BDRRMO interview.

Interview-Identified Gap: Output Clarity for Non-Expert Users

The BDRRMO interviewee raised that users unfamiliar with map conventions may misread susceptibility levels as absolute flood predictions rather than classification-based outputs.

Mitigation: Include a brief legend explanation accessible from the AR view. Add disclaimer on first launch: 'Susceptibility levels indicate the likelihood of flooding based on environmental conditions — not a guarantee of actual flooding.' Brief instructional tooltip accessible during AR use.

2.6 Feasibility Summary

Dimension	Assessment	Verdict
Technical	All features implementable with free, well-documented tools; ML susceptibility classification model trained on open and BDRRMO/CPDCO datasets using scikit-learn/XGBoost; AR rendering via Vuforia ground-plane tracking on flat surfaces — no GPS anchoring required; all screens built in Unity uGUI	FEASIBLE
Financial	Zero licensing cost during thesis period; all tools, and frameworks operate within free tiers sufficient for thesis-scale pilot testing (20–50 participants)	FEASIBLE
Team Capacity	Three-person team with clearly separated AR/UI, ML, and data roles; 10-month phased timeline isolates ML and AR development to prevent bottlenecks	FEASIBLE
User Adoption	Pilot testing achievable through BDRRMO community outreach and direct recruitment from survey respondents; PAGASA-aligned plain-language input removes technical barrier for non-expert users	FEASIBLE
Institutional Support	BDRRMO key informant confirmed openness to data sharing and collaboration; CPDO confirmed on May 18, 2026 that required spatial maps exist and can be released to the research team as hard-copy prints. Barangay boundary maps and historical flood records are available for research use upon formal request.	FEASIBLE
Scope	Tidal flooding and flash floods deferred to future work; vulnerability-adjusted risk scoring deferred to future work; Flood Information Hub and offline mode are stretch	FEASIBLE

Dimension	Assessment	Verdict
	goals only; remaining core features are well-scoped for thesis-level research within 10 months	
Academic Contribution	Novel integration of AR visualization and ML-based flood susceptibility classification for a Philippine LGU context; PAGASA-aligned community input design; BDRRMO-validated problem and concept; correct use of susceptibility (not risk) framing supported by environmental feature set	STRONG